MIS 201 · WEEK 4 · WEDNESDAY 10/14

Lab 4: Inside the Graph

Recap, then the lab, then the debate. Pairs, one machine.

Today

2:00	Project preview, then recap	5 min
2:05	Lab 4, sections 1 to 4	50 min
2:55	**Break**	5 min
3:00	The JAMA argument, in two slides	6 min
3:06	Debate (follow the Gilman Rules!)	37 min
3:43	Wrap-up and attendance	7 min

DUE FRIDAY 10/16, 11:59 P.M.

- Lab 4 writeup, five items
- Quiz on Chapter 4

Answer the writeup items as you go today, and you'll be mostly done by 3:50.

The Prototype project: start thinking now

FINAL PROJECT · 10%

In a team of four, design your own AI system and pitch it. Your spec is the prototype on paper. Building a working version is optional, and so is using AI.

4

people per team

pick your own, or get placed

5 min

the pitch, in class

Nov 30 or Dec 2, four slides max

12/4

paper, slides, reflection

due 11:59 p.m. on Canvas

DUE FRIDAY 10/30, 11:59 P.M.

Scout teammates in today's lab and debate; we'll form teams Monday. Then on Canvas: four names and one paragraph on your idea.

WHERE IT LIVES

prof-califf.github.io/intro-to-ai/prototype-project.html. For ideas, browse the Rundown AI Workflow Hub from Lab 0.

You're graded on whether your design survives the five questions, not on whether you built it.

Recap from Monday

1. **The data trove**

   Records kept for billing and looking things up, big enough to train a model. The owner owns the prediction.

2. **The knowledge graph**

   Dots and labeled lines. A path is a prediction and its reason. A path is a place to look, not a finding.

3. **The data has a shape**

   Thick where money went, thin where it didn't. A low score can mean "no" or "nobody looked."

4. **Who checked?**

   A company's number vs. an outside number. The Epic alert missed 2 of 3 sepsis cases.

5. **Where is the human?**

   In, on, or out of the loop. Every spot has a price. The human gate is a setting you test.

WHAT DO YOU THINK?

Tell the person next to you what a knowledge graph is, in one sentence, and why a low score on a rare disease might mean nobody looked. Go.

Lab 4: today you build a graph, stand at the gate, and read Every Cure's real scores with Claude. Then the room debates who should sign off.

Part 1: the lab

50 MIN · PAIRS

1	**Build a knowledge graph, ask it for a cure**	15 min	done by 2:20
2	**You're the doctor in the loop**	10 min	done by 2:30
3	**Ask Claude what Every Cure should study**	20 min	done by 2:50
4	**Under the hood**	5 min	done by 2:55

Section 1 is section 2.2's graph. Section 2 is 2.5's gate. Section 3 is 2.3's gap, in Every Cure's own data.

HOW IT WORKS

- Pairs, one machine.
- Sections in order; steps match the buttons.
- Answer the writeup items as you go.
- Section 3: don't type anything about your own health.
- Log out of Claude when section 3 is done.

Section 3: the real scores

LAB · SECTION 3

Every Cure publishes MATRIX's scores for about 39.5 million drug and disease pairs. Today you choose a disease, ask Claude which two drugs Every Cure should look at first, and check its answer.

9,323

facts on myeloma

thickest record, 43 known treatments

39

facts on iMCD

Fajgenbaum's disease, none known

994th

siltuximab for iMCD

FDA-approved for it in 2014

WATCH FOR

Claude's message looks ready to send. Is it? Then: compare POEMS to myeloma, check one study, ask about a low score, build a chatbot.

THE REVEAL

Joseph's drugs: cyclophosphamide ranks 2nd, carfilzomib 194th. Siltuximab, approved for iMCD: 994th.

A low score can mean "no," or "nobody wrote it down under this name."

BREAK · 5 MINUTES

Back in five

When you come back: the JAMA argument in two slides, the Gilman Street rules, then your side. Left half of the room is Side A, right half is Side B.

The JAMA argument: take the doctor out

DEBATE SETUP

THE CLAIM · JAMA, AUGUST 2026

AI alone will soon beat doctors.

And beat doctors working with AI, at the thinking work: taking a history, diagnosing, choosing tests, recommending treatment, managing chronic disease.

In the authors' words:
"AI-alone medical care is likely to be better than physician-only or physician-AI hybrid care."

THE ASK

Regulators drop the human-in-the-loop default.

The FDA, state medical boards, Medicare and insurers all assume a licensed person makes the final call.

WHO'S SAYING THAT?

Ezekiel Emanuel (oncologist, bioethicist, Penn) and Abe Baker-Butler (research fellow). Neal Khosla, CEO of Curai, an AI health company. Vinod Khosla, an investor in Curai. They disclosed it.

Imagine it's five years out and they're right. Should the rule that puts a doctor at the gate stay?

The study behind it

DEBATE SETUP

Goh and colleagues, JAMA Network Open, 2024. 50 doctors, up to six hard real patient cases written up as summaries, one hour. Half could use GPT-4. Experts graded each doctor's reasoning on a scoresheet.

74%

Doctors alone

no AI

76%

Doctors using GPT-4

in the loop

~90%

GPT-4 alone

out of the loop

WHAT SIDE A WILL SAY

Doctors with GPT-4 scored 76; GPT-4 alone scored 90. They used it like Google and ignored it when it disagreed. Those mistakes are ours, not the machine's.

WHAT SIDE B WILL SAY

These were cases on paper, already boiled down by a doctor. The study's own authors said not to read it as a case for AI alone.

Same three numbers, two readings. That's the debate.

Debate rules: the Gilman Street rules

APPLY THESE TODAY

Same rules as always, borrowed from 924 Gilman Street. Here's what each one means in a debate.

NO RACISM · NO SEXISM · NO HOMOPHOBIA · NO TRANSPHOBIA	Argue with ideas, never at people.
NO FIGHTING	No attacking a position without evidence. "That's wrong" isn't a point. "That's wrong, because the 90 was scored on cases a doctor had already boiled down" is.
NO STAGEDIVING	No interrupting, piling on, or ambushing. One speaker at a time; let them finish.
NO MAJOR LABEL BANDS	No blind love or hate for an AI company or a profession. Curai, the AMA, doctors, AI: give reasons, not fandom.

The debate: who signs off on a diagnosis?

37 MIN

Imagine it's 2031. AI gets diagnoses right 90% of the time or better, with no doctor involved.

In that future, should regulators stop requiring a doctor's sign-off?

SIDE A · LEFT HALF OF THE ROOM

Yes: AI alone

Drop the rule. The AI makes the diagnosis and no doctor has to sign off. Test it the way we test a drug. Start from JAMA, the 74, 76, 90, and the Goh doctors who ignored the AI.

Cost to name: when it's wrong, nobody is in the room, and the best evidence so far is cases on paper.

SIDE B · RIGHT HALF OF THE ROOM

No: doctor with AI

Keep the rule. A doctor who knows how to use AI works with it and signs every diagnosis. Start from Wachter, Every Cure's gate, and the Goh doctors who used AI like Google: a training problem.

Cost to name: on the best numbers we have, doctors with AI scored 76 and AI alone about 90, and every signature is doctor time patients wait for.

Resolve it: the three trials put the human in three places and got three answers. Name one task where each side would move the doctor, how it should be tested first, and who answers when it's wrong.

Your evidence: four cards, one per number

1 The JAMA viewpoint and the study

The claim, the 74 / 76 / 90, and who wrote it. Doctors used GPT-4 like Google and ignored it when it disagreed. The study's authors cautioned against AI alone.

2 The pushback

Six doctors' groups incl. the AMA: mostly simulated cases. Wachter: in an Oxford study, people using the AI got the right condition only about a third of the time. The doorman fallacy.

3 Three hospital trials

MASAI: radiologist on the loop, 29% more cancers, 44% less reading. Fetal ultrasound: AI live beat AI afterward. PROMETHEUS: faster scans, no confirmed accuracy change.

4 Every Cure's human gate

A score is a claim about the record. People read the ranking; doctors who knew Joseph made the call. A high score can also become cover for a risky decision.

Every side gets all four. If yours seems to help the other side, find what it means for yours.

How the debate runs: prep

15 MIN

1 Get your side — 2 min

Left half is Side A, right half is Side B. Groups of 2 or 3 on your side. Count off 1 to 4: that's your evidence card.

2 Push your thinking with Claude — 10 min

Send the sample prompt under your card. Then tell it to argue against you, hard. If it cites a study you'll use, look it up before you use it. Close it, log out, write notes in your own words.

3 Team up and pick speakers — 3 min

Join one or two other groups on your side, so your team is 6 or 7: three teams per side. Pick your single best point, an opener, and a rebutter.

YOUR NOTES

1. Your strongest point, in one sentence
2. The specific evidence behind it
3. The honest cost of your position
4. What your side would rather you didn't notice

THE RULE ON AI

Use it to research. Don't read from it.

How the debate runs: the rounds

22 MIN

Round 1	**Openings** — Openers to the front: A1, A2, A3, then B1, B2, B3. About 45 seconds each: your claim, your evidence, the cost of your plan.	6 min
	Huddle — Back to your team. Rebutters pick one point the other side made.	2 min
Round 2	**Rebuttals** — Side B first this time, then A. Answer one specific point, with evidence. No new arguments.	6 min
Round 3	**Switch** — 30 seconds with your team: what was the other side's best point? Then someone new argues it in one sentence.	4 min
Recap	**What would change your mind?** — Out loud. Name the specific thing you would have to see.	4 min

THE GILMAN RULES, APPLIED

One speaker at a time. Let them finish.

Every point needs evidence.

Answer the argument, not the person.

Reasons, not fandom.

Whoever names real evidence that would move them wins, whichever side they were handed.

BEFORE YOU LEAVE

Upload your notes

Attendance = your notes from today, uploaded to Canvas right after class.

At home: only the writeup. Five items, what you saw and did today as evidence. Item 2 is the doctor in the loop; item 4 is the debate; item 5 is your commitment card.

Friday 10/16, 11:59 p.m.: Lab 4 writeup and the Chapter 4 quiz, both on Canvas.

Log out of Claude before you go. Shared machines.